

SMARTBRANCH 360

Fault Cards - Symptom, Root Cause, Fix and Verification

Fault 1 - VLAN 20 missing from trunk

Field	Details
Intentional change	Remove VLAN 20 from SW2 Fa0/24 allowed VLANs.
Symptom	Guest laptops lose connectivity and/or DHCP.
Root cause	VLAN 20 is not allowed across the SW2-SW1 trunk.
Evidence	show interfaces trunk; Python trunk check reports missing VLAN 20.
Fix	Restore: switchport trunk allowed vlan 10,20,30,99.
Verification	Renew Guest DHCP and ping 10.10.20.1.

Fault 2 - Wrong Guest DHCP network

Field	Details
Intentional change	Change GUEST DHCP pool to 10.10.30.0/24.
Symptom	Guest clients receive incorrect/no usable addressing.
Root cause	The DHCP pool does not match Guest VLAN 20.
Evidence	show ip dhcp pool; Python DHCP check reports mismatch.
Fix	Restore network 10.10.20.0 255.255.255.0 and gateway 10.10.20.1.
Verification	Renew DHCP; confirm 10.10.20.x and ping gateway.

Fault 3 - Guest ACL blocks DNS

Field	Details
Intentional change	Remove the UDP/53 permit to Server0.
Symptom	Guest IP connectivity remains, but hostname/DNS resolution fails.
Root cause	Guest ACL denies DNS traffic to 10.10.30.10.
Evidence	show access-lists GUEST-ACL; Python ACL check reports missing DNS rule.
Fix	Restore permit udp 10.10.20.0 0.0.0.255 host 10.10.30.10 eq 53.
Verification	Retry DNS lookup; Python ACL check returns PASS.

Fault 4 - Wrong VLAN 20 gateway IP

Field	Details
Intentional change	Change R1 G0/1.20 to 10.10.30.1.
Symptom	Guest routing no longer matches the intended subnet.
Root cause	VLAN 20 subinterface has the Server VLAN gateway

	address.
Evidence	show ip interface brief; Python interface check reports mismatch.
Fix	Restore G0/1.20 to 10.10.20.1 with dot1Q 20.
Verification	Confirm G0/1.20 is 10.10.20.1 up/up and ping it from Guest.

Fault 5 - NAT overload rule missing

Field	Details
Intentional change	Remove the PAT/NAT overload statement.
Symptom	Internet-bound traffic is not translated when a valid upstream exists.
Root cause	The NAT overload rule is absent.
Evidence	show running-config include ip nat; Python NAT check reports missing rule.
Fix	Restore ip nat inside source list 1 interface GigabitEthernet0/0 overload.
Verification	Run NAT check and, with an upstream simulator, verify translations.

Demo method: introduce one fault at a time, capture the Cisco show output, run the Python checker, explain the finding, apply the fix, and verify recovery. Restore the healthy baseline before moving to the next fault.